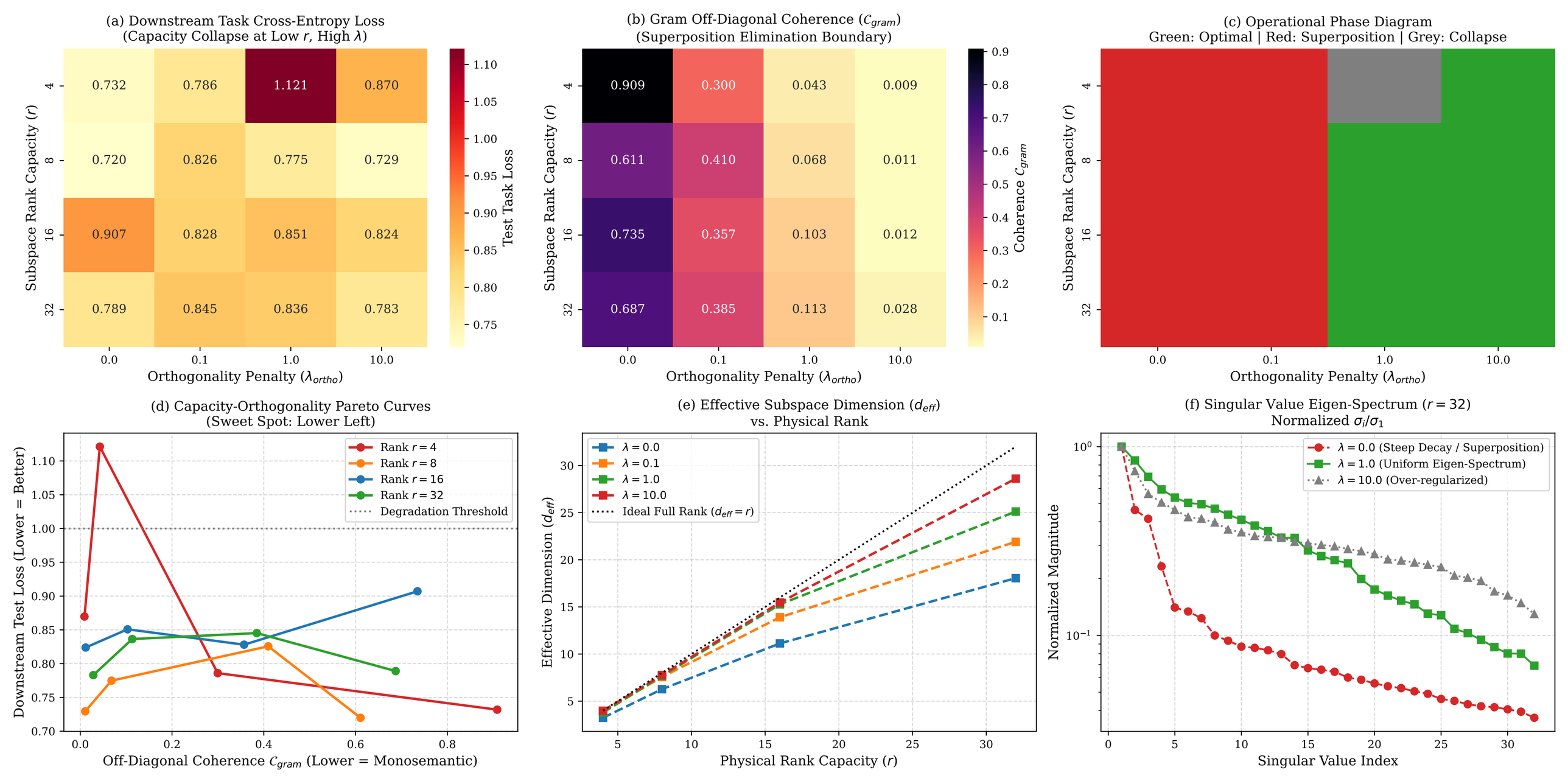
(a) Downstream Task Cross-Entropy Loss
(Capacity Collapse at Low r, High λ)
Subspace Rank Capacity (r)
Orthogonality Penalty (λortho)
Test Task Loss
0.732 0.786 1.121 0.870
0.720 0.826 0.775 0.729
0.907 0.828 0.851 0.824
0.789 0.845 0.836 0.783
(b) Gram Off-Diagonal Coherence (Cgram)
(Superposition Elimination Boundary)
Coherence Cgram
0.909 0.300 0.043 0.009
0.611 0.410 0.068 0.011
0.735 0.357 0.103 0.012
0.687 0.385 0.113 0.028
(c) Operational Phase Diagram
Green: Optimal | Red: Superposition | Grey: Collapse
(d) Capacity-Orthogonality Pareto Curves
(Sweet Spot: Lower Left)
Rank r = 4
Rank r = 8
Rank r = 16
Rank r = 32
Degradation Threshold
Downstream Test Loss (Lower = Better)
Off-Diagonal Coherence Cgram (Lower = Monosemantic)
(e) Effective Subspace Dimension (deff)
vs. Physical Rank
λ = 0.0
λ = 0.1
λ = 1.0
λ = 10.0
Ideal Full Rank (deff = r)
Effective Dimension (deff)
Physical Rank Capacity (r)
(f) Singular Value Eigen-Spectrum (r = 32)
Normalized σi/σ1
λ = 0.0 (Steep Decay / Superposition)
λ = 1.0 (Uniform Eigen-Spectrum)
λ = 10.0 (Over-regularized)
Normalized Magnitude
Singular Value Index